

Shivang Arya

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EDUCATION

Hemvati Nandan Bahuguna Garhwal University

Bachelor of Computer Science, CGPA: 7.59

Dehradun, Uttarakhand

Aug. 2023 – June 2026

National Institute of Technology, Warangal

Master of Computer Applications

Warangal, Telangana

Aug. 2026 – July 2029

EXPERIENCE

Technical Contributor, GigaVector (Open Source)

Feb. 2026 – Apr. 2026

Vector Database

Remote

- Patched missing wrapper bindings in `python_compat.c` that caused MSVC exported DLL builds to crash with `AttributeError` failures when Python clients accessed namespace management APIs
- Restored stability for Windows test targets by completing native to Python wrapper exposure paths, while also refactoring header organization and formatting for maintainability

PROJECTS

tinyweb | *High-Performance Web Server* | C, POSIX Sockets, pthreads

- Built a modular HTTP/1.1 web server from scratch in raw C using POSIX sockets, implementing low-level TCP networking without external frameworks
- Designed a multi-threaded request execution pipeline using a condition variable based thread pool for concurrent client connection handling
- Implemented a variadic routing engine mapping HTTP methods and URIs directly to function pointer callbacks through custom hash map lookups
- Developed custom memory managed data structures including linked lists, queues, binary trees, and dictionaries to support asynchronous request processing

strandy | *Concurrent Task Executor* | C, pthreads, Concurrency

- Engineered a lightweight POSIX compliant thread pool library in raw C for asynchronous task execution using condition variables and mutex synchronization
- Implemented a thread-safe FIFO work queue supporting concurrent producer-consumer scheduling without spin lock CPU overhead
- Built synchronization primitives around pending and active worker state tracking to safely coordinate executor shutdown and workload completion

rvlib | *Runtime Generic Collections* | C, Generic Data Structures

- Architected a generic container library in C using descriptor-based runtime polymorphism to provide reusable data structures without language-level generics
- Implemented linked lists, stacks, queues, dequeues, and a chained hash table with deep-copy insertion, custom hashing strategies, and type-aware element management
- Built an extensible type descriptor framework exposing pluggable comparison, hashing, copy, print, and destruction routines for user-defined data models
- Designed deterministic memory ownership semantics and recursive cleanup paths to safely manage dynamically allocated resources across nested container hierarchies

TECHNICAL SKILLS

Languages: C, Python, JavaScript, Lua, SQLite, PostgreSQL

Systems: POSIX APIs, Socket Programming, Multithreading, Memory Allocators

Frameworks: Flask, Astro.js

Developer Tools: Git, Docker, GDB, Valgrind, NeoVim, Arch Linux